Improved ResNet18: An Interpretable Deep Learning Framework for Automated Eye-Disease Classification

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***Abstract*—Early and accurate detection of eye diseases is essential, as observational methods such as fundus examination are time-consuming and prone to human error. In response, we propose an effective deep learning framework for multi-class eye disease classification using simple eye images. The framework compares various architectures, such as CNNs and vision transformers and finds ResNet18 the best one. The steps involved in preprocessing are grayscale conversion, CLAHE, and Gaussian filtering, as well as data augmentation and class balancing with SMOTE. ResNet18 with test accuracy of 97.44% macro sensitivity of 95.11% and macro specificity of 99.36% and lowest training and inference time among models. The Grad-CAM visualization demonstrated that the model concentrates on clinically significance areas within the eye which improve interpretability. The strength and statistical reliability of the framework were confirmed by extensive testing with 5-fold cross- validation, confidence intervals, and using the McNemar test. Such findings demonstrate that the proposed system is correct, effective, and interpretable and has a high potential of being an automated diagnostic tool to classify eye disease.**

***Index Terms*—Medical image classification, Deep learning, CNN, Vision Transformer, Grad-CAM, Eye disease detection**

I. INTRODUCTION

Eyes are the most crucial and sensitive organ of human structure. This important organ can be affected with multiple diseases, which can lead to permanent damage to the retina. Visual impairment, blindness can affect daily human activities such as reading, walking, recognizing faces [1]. Cataract, eyelid disorder, conjunctivitis, and uveitis are some eye- diseases that can cause inflammation, redness of the eye and blurry vision. Prolonged exposure to the sunlight can cluster the protein of the eye lens together, results to impaired vision. This process often leads to the development of cataracts. Aging is also a factor of cataract formation [2]. Conjunctivitis, also known as pink eye, causes due to inflammation of conjunctiva [3], [4]. Blindness in 5-20% of cases in the US and Europe, 25% of cases developing countries is directly associated by uveitis [5].

For classifying ophthalmic diseases, doctors typically rely on observational method, which is often time-consuming and susceptible to human error [6]. High cost, lack of awareness, shortages of ophthalmologists often hinder people from taking necessary precautions [7]. These issues imply that ophthalmology has become a prominent sector where application of Deep Learning is crucial. With a proper application of Artificial Intelligence and Deep Learning, these issues can be addressed effectively, where models can classify diseases.

Despite all advancements, the application of transformers and lightweight CNNs for eye disorder recognition using standard eye- images remains sufficiently unexplored. Existing study often emphasize on Fundus or slit-lamp images. Also, many works do not sufficiently address clinical interpretability, which is essential for clinical deployment. Inspired by these limitations, this study introduces a unified framework for multi-class classification using CNN, hybrid and transformer architectures. This study focuses on a deep learning- based eye-disorder classification using simple eye images. CNN can automatically detect prominent attributes without human supervision [8]. Transformer models, on the other hand enables direct interactions among different image regions [9], while hybrid models combine both CNNs and Transformers. Many studies, for example, Babaqi et al.

[7] have focused on only CNN and Transformer-based models for classification. Some researches, for instance, paper [10] are limited only a particular eye-disorder. In our research, we propose a CNN, hybrid, and transformer-based framework to automate eye disease classification from color eye-images.

The primary contributions of this work can be summarized in the following way. To begin with, a single deep learning architecture is introduced to classify multi-class eye diseases on simple eye images in RGB format and apply CNN, transformer and hybrid frameworks with a unified training and testing process. Second, a proper pre-processing and data balancing methodology that will be based on image enhancement, data augmentation, and SMOTE are presented to enhance feature discrimination and solve the issue of the class imbalance. Third, interpretation of models is realized by explainable AI analysis with Grad-CAM, which can allow visual verification of the clinically relevant areas of decision-making. Lastly, the strength and stability of the proposed framework are authenticated through cross-validation and a rigorous statistical analysis, which determine its appropriateness in automated screening of ocular diseases.

II. LITERATURE REVIEW

The use of machine learning on tabular data and medical images has grown rapidly, particularly for eye disease

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diagnosis[11], [12], [13], [14]. These approaches improve accuracy, reduce diagnostic time, and support clinical decisions. This section reviews key studies on automated eye disease classification using both imaging and hybrid methods.

A. *Fundus-Based Approaches*

Nakayama et al. [5] introduced CAD-EYE, a hybrid deep learning model for the classification of eye diseases. The CAD-EYE model is a combination of MobileNetV2 and EfficientNetB0 with feature fusion and uses an XGBoost classifier. The model used global average pooling and dense layers with ReLU and SoftMax activation functions and trained using the Adam optimiser and cross-entropy function for 10 epochs. Accuracy, F1-score, recall, and precision were used for evaluation, and CAD-EYE reported 98% accuracy, which outperformed other models, including VGG16, ResNet50, and InceptionV3. Jain et al. [15] proposed the Adaptive Hybrid Net- CNN: a binary & multi-class eye disease detection using a deep hybrid model with InceptionV3, MobileNetV3-Small, ResNet-101, & Adaptive CNN layers. The binary dataset split is train/test = 80/20 while multi-class datasets split into three classes & SMOTE. The evaluation metrics used were acc, F1-score, precision, recall, sensitivity & specificity. The model resulted in a higher accuracy rate & sensitivity with 98.45% & 96.36%, respectively.

A. *OCT-Based Approaches*

Zhou et al. [16] proposed a ViT-based model with multiclass classification of retinal diseases using OCT images with NORMAL, CNV, DME, and DRUSEN classes. Features were learned in this model using image patching, token embedding involving positional encoding, and multi-head self-attention. It was evaluated on the basis of accuracy, sensitivity, specificity, and confusion matrices with 96.57% validation accuracy and 95.76% test accuracy over EfficientNet, ResNet50, and VGG16 models. Akc¸ a et al.[17] performed a comparative analysis of vision transformer architectures for multiclass classification of retinal diseases from OCT images using transformers. A balanced subset of the Labelled OCT dataset with CNV, DME, Drusen, and Normal classes was utilised. Among the attention- driven models, Vision Transformer (ViT), T2T-ViT, and Mobile-ViT were tested using the AdamW optimizer with cross-entropy loss and standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Among all models, the highest accuracy of 99.17% was obtained using Mobile-ViT. Hemalakshmi et al. [18] developed a lightweight hybrid deep learning model called SViT for accurate classification of retinal OCT images. The proposed approach applies both SqueezeNet for local feature extraction and the Vision Transformer (ViT) for the handling of global relationships in the image to support both two-class classification (normal and retinal disease) and multiclass classification (normal, DME, CNV, and Drusen) of the retinal disease. The SViT model achieved an initial validation accuracy of 90.27% and performed very well in both binary and multiclass classification tasks. The drawback of the paper is the potential large computational cost required in the training process of the proposed hybrid model. Lin et al. [19] employed a Two-StreamCNN-Transformer architecture for the classification of retinal disease. Spatial features were modeled by CNNs, and transformers interpreted overall information. The Adam optimiser and cross-entropy loss function, data augmentation techniques of rotation, flip, cropping, and adding Gaussian noise were employed. Accuracy of 96%, F1-score of 0.96, and an AUC of 1.00 were achieved by LGSF-Net, outperforming ResNet50, ViT, Swin, and InceptionV3 models while utilizing lower parameters of 18.7K and FLOPs of 0.93G.

III. Methodology

In this section, we have discussed the methods used for the classification of eye-disease using simple RGB eye images. The overall approach is demonstrated in Fig. 1.

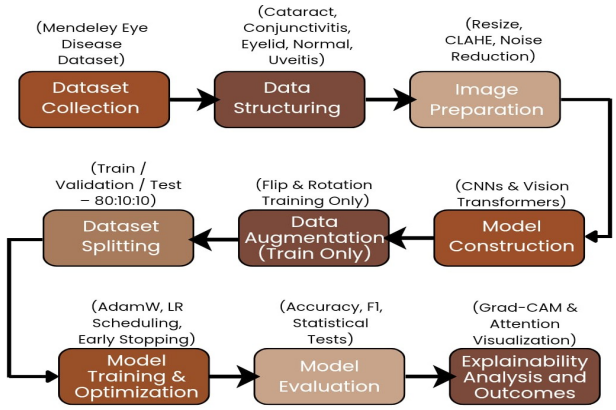


Fig. 1: Overall workflow of the proposed eye disease classification framework.

A. *Dataset Description*

In the study of the eye-disease classification, we used a publicly available dataset on the Mendeley Data repository [20], which is directly aimed at allowing automated detection of eye diseases using an analysis of simple eye images. This dataset consists of real clinical triadic eye images of five classes; Cataract, Conjunctivitis, Eyelid, Normal and Uveitis. Particularly, it contains 544 images of Cataract, 357 of Conjunctivitis, 525 of Eyelid, 649 of Normal, and 223 of Uveitis, which is a balanced and rich representation that enables good training and strong evaluation of the classification models. Figure 2 shows sample images of each class from the dataset.

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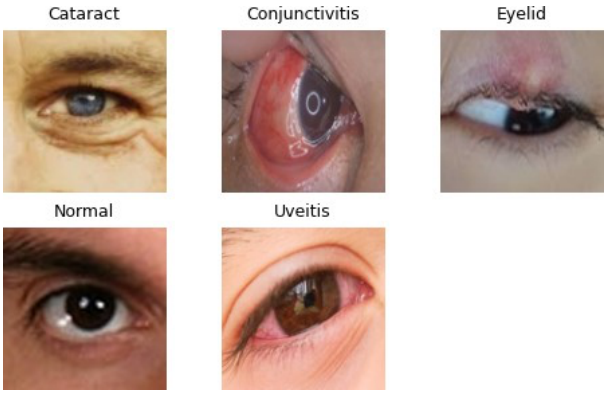


Fig. 2: Sample images per class from the dataset.

B. *Data Preprocessing*

A preprocessing pipeline was applied to all raw images for image quality enhancement and noise suppression to ensure the output is compatible with deep learning models. Since OpenCV uses the BGR channel order, BGR to RGB conversion was performed for overall consistency in operation. For stable training, all images were resized to 224×224 pixels. After resizing, the RGB images were transformed to grayscale to reduce color dependency. To boost local contrast and visibility of minor features, Contrast Limited Adaptive Histogram Equalization-CLAHE was implemented on the grayscale images, configured with a clip threshold of 2.0 and a grid partition of 8x8 tiles. Following contrast enhancement, a Gaussian filtering using 3x3 convolution kernel was employed to minimize high-frequency noises.

C. *Dataset Splitting and Augmentation*

Post-preprocessing stage, the dataset was partitioned into train, validation and test subsets to support stable model learning and fair model evaluation. A split directory was created to prevent data leakage from earlier data. Images were randomly shuffled and partitioned using 80:10:10 training, validation and testing ratio, applied independently on each class. Table I summarizes the image counts for each dataset split.

TABLE I: Train/Validation/Test split per class.

Class	Train	Val	Test
Cataract	435	54	55
Conjunctivitis	285	35	37
Eyelid	420	52	53
Normal	519	64	66
Uveitis	178	22	23

Data augmentation was exclusively performed on the training set only to promote generalization while preventing overfitting effects. Inputs were uniformly rescaled to 224x224 and subjected to stochastic transformations by random horizontal flipping and minor angular rotations (±10°).

D. *Model Architectures and Training Strategy*

1) ***Evaluated CNN and Transformer Models****:* Six architectures, namely, ResNet18, Swin-Tiny, DeiT-Tiny, LeViT-256, MobileViT- XS, and ShuffleNetV2, were systematically evaluated in this study to classify eye disease from simple eye images. In particular, Swin- Tiny, a hierarchical vision transformer, is a self-attention vision transformer with a hierarchical structure, and the local windows are overlapped. It is fine-tuned with a custom classification head, and dropout to reduce overfitting. In the same light, DeiT-Tiny, which is a data-efficient transformer, was pre-trained and fine-tuned to determine the viability of small transformers on small data. Also, LeViT-256 is a CNN-Transformer hybrid which uses convolutional bias and lightweight self-attention layers, and uses a modified fully connected layer and early stopping to improve generalization. In addition, MobileViT-XS is based on combining the light convolutions with transformer modules to effectively extract local and global features, using the same preprocessing and training pipeline. Lastly, ShuffleNetV2 is a small CNN, using channel shuffling and pointwise grouped convolutions, which was used as a baseline with a modified output head to classify five types of diseases.

Proposed Improved ResNet18 Model*:* Among all evaluated architectures, ResNet18 consistently outperformed other models' performances reflecting robust generalization under all reported metrices. The model follows residual learning framework, where identity shortcut connections enable stable gradient propagation across deep network layers. This characteristic is highly suited for medical image analysis, where subtle textural patterns and detailed visual cues must be retained. In this study, the ImageNet-pretrained ResNet18 was modified by replacing the final FC layer with a regularized linear classification head to limit model overfitting. Label smoothing was also introduced within cross-entropy loss formulation to reduce over- confident predictions and applied AdamW optimization with weight decay to promote model generalization. Later on, gradient magnitude clipping and adaptive learning rate scheduling were employed to ensure stable training. The combination of robust CNN inductive biases, strong regularization techniques and transfer-based optimization outperform both CNNs and transformer model architectures on the eye-disease prediction task. Fig. 3 illustrates the architecture of ResNet18-CNN Model.

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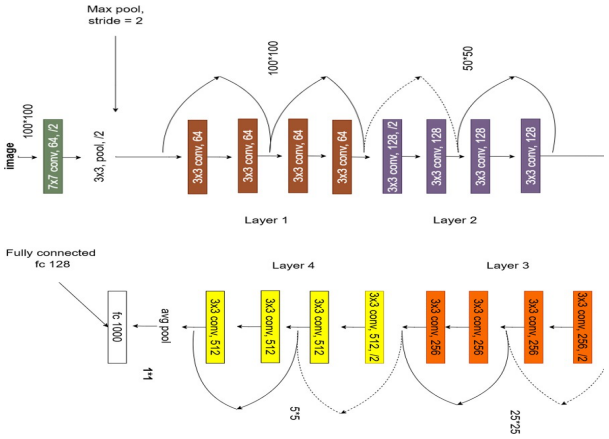


Fig. 3: Improved ResNet18- a CNN-based Model Architecture

E. ***Data Training Strategy***

All the experiments were performed by using the PyTorch deep learning framework on a Kaggle Notebook setup with NVIDIA Tesla P100 GPU. All the tested models were trained using the same standardized training configuration, where augmented, rescaled, and normalized input images of dimension $224 \times 224 \times 3$ were employed in the five-class classification task with AdamW (learning rate 3×10^{-4}, weight decay 1×10^{-4}) and CrossEntropyLoss (label smoothing 0.1) as the loss functions, and the batch size was 32, the dropout rate was 0.5, and the gradient clipping was 5.0. The maximum amounts of epochs trained were 40 and early stopping (patience = 5) and ReduceLROnPlateau (factor = 0.5, patience = 2), which ensured fair, controlled and unbiased comparison of all the architectures considered.

F. ***Evaluation Metrics and Statistical Tests***

Model performance was measured using both traditional and domain-specific metrices. In addition to total accuracy, class-wise evaluation metrices including Sensitivity, Specificity, Positive Predictive Value (PPV), and Negative Predictive Value (NPV) were de- rived to examine diagnosis performance. For classification accuracy, bootstrap-based Confidence Interval (CI) was computer using iterative resampling of the test set. For statistical comparison, McNemar statistical test was performed to evaluate paired prediction. It confirms whether performance variations are significantly meaningful.

G. ***Explainable AI using Grad-CAM***

For better understanding of the proposed models, Gradient Weighted Class Activation mapping (Grad-CAM) was generated for top-performing model, ResNet18. Grad-CAM produces category- focused heatmaps by analyzing gradients into the final feature extraction layers, highlighting the most relevant regions for classification results. This process allows visual inspection of whether the networks focus on medically relevant eye regions instead or irrelevant non- informative background.

IV. Results and Discussion

In this section, the performance of each of the evaluated models is presented and interpreted, with the important distinctions in terms of accuracy, sensitivity, and specificity to automated eye disease classification being highlighted. The findings of these experiments provide an understanding of the strengths and weaknesses in practice of each architecture when applied to the real-world ophthalmic image.

A. ***Model Performance Comparison***

TABLE II: Performance comparison of different models on training, validation, and test sets.

Model	Train Acc.	Validation Acc.	Test Acc.
ResNet18	1.0000	0.9604	0.9744
Swin-Tiny	0.9951	0.9559	0.9444
DeiT-Tiny	0.9858	0.9604	0.9359
LeViT-256	0.9989	0.9780	0.9402
ShuffleNetV2	0.9989	0.9648	0.9359
MobileViT-XS	0.9940	0.9648	0.9573

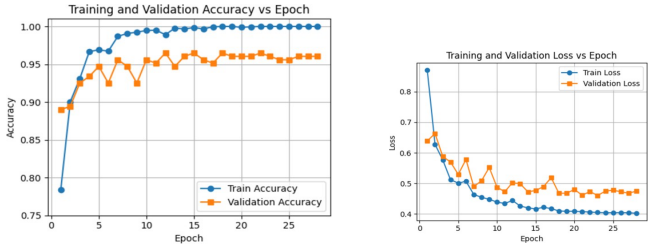
Table II reports performance across all data splits (training, validation, test set) of all considered architectures. Among the six evaluated networks, ResNet18 obtained the highest testing accuracy of 97.44%, indicating stronger generalization ability. Although the model has near-saturated training results, the minor performance gap between training, validation (96.04), and test accuracy confirms that overfitting was well controlled by means of explicit regularization and early stopping. LeVit-256 attained high validation accuracy (97.80%) but showed a noticeable drop in test performance pointing to moder-ate overfitting. ShuffleNetV2 and DeiT-Tiny performed well during training but had lower generalization on test data (93.59%) indicating that compact architectures may have difficulty capturing subtle ocular features with restricted dataset sizes. Vision transformer-based models such as Swin-Tiny and DeiT-tiny showed strong training performance but reflected relatively lower accuracy during evaluation and testing, highlighting shortcomings in extracting class-specific patterns from existing dataset scale. The MobileViT-XS hybrid model outperformed transformer-only models, achieving a 95.73% accuracy on test set, demonstrating the benefits of incorporating convolutional feature biases with transformer attention layers. Overall analysis shows CNN architectures delivered more stable and consistent accuracy across all data splits, making ResNet18 the most robust model for ocular disease classification.

B. ***Accuracy and Loss Curve Analysis***

Fig. 4 shows the training and validation accuracy curve against epochs of the best performing model (ResNet18) exhibit early learning convergence during early epochs, with training accuracy crossing 95% in the initial training iterations because of ImageNet pretrained weights. As training progress, the model attains almost perfect training

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accuracy while preserving stable validation performance around 96-97%. The tight alignment between training and validation curve underscores the effectiveness of stabilization techniques. Similarly, the loss curve exhibits a steady decrease with training loss stabilizing around 0.40 and validation loss near 0.46. No instability or sudden increase in validation loss is noticed, confirming convergence stability and controlled learning behavior. The ultimate test accuracy of 97.44% closely aligns with validation accuracy, supporting the robustness of the optimized model on unseen data.



(a) Accuracy curve. (b) Loss curve.

Fig. 4: Training and validation learning curves

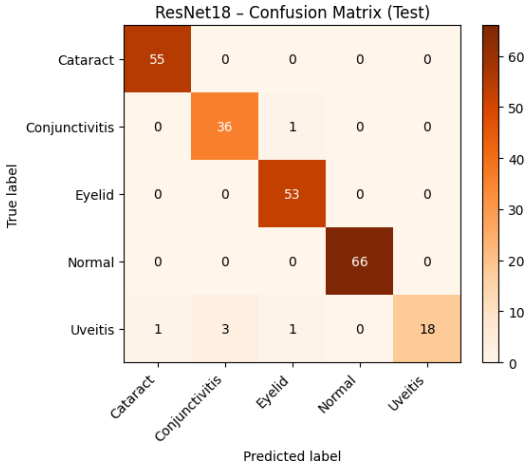


Fig. 5: Confusion Matrix for best performing model (ResNet-18).

Fig. 5 illustrates the confusion matrix visualization for the best performing model - ResNet18 analyzed on the test dataset. It shows highly accurate prediction for cataract, eyelid, and Normal classes with zero misclassification cases, reflecting strong class separation. The majority of errors for ResNet18 are concentrated in the Uveitis samples, where a few instances are predicted incorrectly. 3 samples of Uveitis are wrongly predicted as Conjunctivitis which shows a minor contradiction between Conjunctivitis and Uveitis classes. It might happen in case of visual resemblance among inflammatory ocular conditions. However, the prediction model maintains strong diagonal dominance confirming reliable multi-class classification, while identifying Uveitis as the most error-prone class due to inter- class visual overlap.

D. Evaluation Metrics Comparison

TABLE III: Comparison of test accuracy, confidence intervals, and macro-averaged medical evaluation metrics across models

Model	***CI**^{Low}* *95%*	***CI**^{High}* *95%*	***Sensitivity***	***Specificity***	***PPV***
ResNet18	0.9530	0.9915	0.95110.9936	0.9738	0.9530
Mobile ViT-Xs	0.9316	0.9829	0.9378	0.9898	0.9379
Swin-Tiny	0.9145	0.9744	0.9249	0.9869	0.9202
LeViT-256	0.9103	0.9701	0.9212	0.9852	0.9296
DeiT-Tiny	0.9060	0.9658	0.9347	0.9851	0.9135
ShuffleNetV2	0.9060	0.9658	0.9211	0.9847	0.9163

Table III reports detailed comparison of all trained architectures using diagnostic evaluation metrices, including 95% confidence intervals (CI), macro-averaged sensitivity, specificity, PPV and NPV. Among all, ResNet18 outperformed all other models on the test split (97.44%), with the tightest CI (95.30%-99.15%), indicating robust generalization and statistical stability. ResNet18 also maintained the strongest balance between sensitivity (95.11%) and specificity (99.36%), which is crucial for limiting both false-negative and false- positive errors in clinical diagnosis. MobileViT-XS achieved second- best performance, particularly in specificity (98.98%) and NPV (98.98%), confirming its robustness despite its parameter-efficient design. Vision transformer models such as Swin-Tiny, LeViT-256, and DeiT-Tiny delivered competitive but slightly lower results, likely due to their strong data dependency. While, ShuffleNetV2 showed efficient computation but modestly decreased sensitivity and PPV, indicating increased misclassification risk in less frequent disease categories.

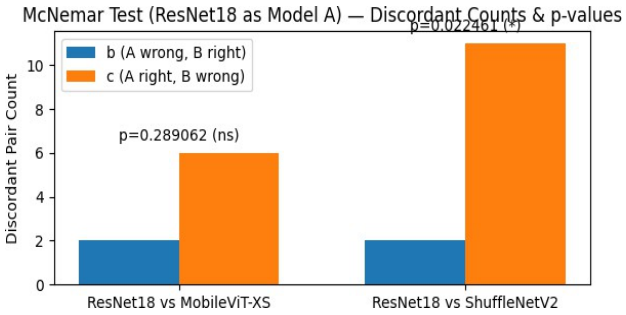


Fig. 6: McNemar paired test to evaluate the statistical significance of the best model.

To systematically assess whether the performance gaps were statistically justified, the McNemar paired test was performed using test-set outputs as observed in Fig. 6. The comparison between ResNet18 and MobileViT-XS produced a p-value of 0.289, indicating equivalent statistical

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performance, even though ResNet18 attained overall higher accuracy. In comparison, the test between ResNet18 and ShuffleNetV2 returned a statistically significant p-value of 0.022, validating a significant performance improvement in the best model. These findings clearly justify ResNet18 as a stable and top- performing architecture, especially relative to lightweight CNN mod- els, while preserving competitive performance against transformer- driven approaches. Fig. 6 represents The McNemar paired test results comparing ResNet18 with MobileViT-XS and ShuffleNetV2 by showcasing the distribution of discordant prediction pairs and their corresponding p-values.

E. 5-Fold Cross-Validation Result

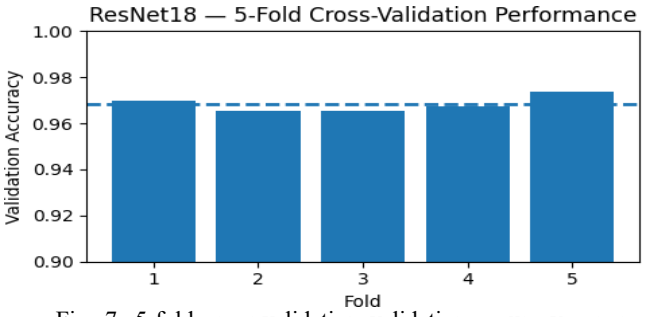


Fig. 7: 5-fold cross-validation validation accuracy across folds (ResNet18).

To further assess the consistency and generalization ability of the proposed pipeline, a 5-fold cross-validation experiment was introduced using the optimized model ResNet18. Fig. 7 shows the gradual cross-validation accuracy of the best model against each fold. The dataset was segmented into five non-overlapping folds, where each fold was used as validation in one iteration while the remaining four folds were assigned to training. The validation accuracy across five folds stayed uniformly high, ranging roughly between 96.5 and 97.4%, indicating stable training dynamics. The small dispersion across fold results demonstrates that the model is insensitive to data splits. This consistency validates that the learned visual features are consistent and generalizable.

F. Training and Inference Time Analysis

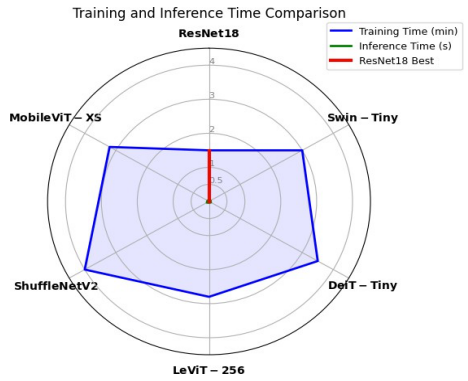


Fig. 8: Comparison of training and inference efficiency across all evaluated models, highlighting ResNet18 as the fastest and most computationally efficient architecture.

Fig. 8 demonstrates how training and inference of all the compared models can be relatively efficient. It is evident that ResNet18 is the fastest due to its simplicity in computing and its optimized architecture. Transformer-based models like Swin-Tiny, DeiT-Tiny and MobileViT-XS, as well as hybrid CNN models such as LeViT- 256 are slower because their attention mechanisms are complicated. Although lightweight, ShuffleNetV2 demonstrates middle performance in relation to ResNet18. Generally, the present comparison has illustrated the trade-off between model complexity and cost of computation, with ResNet18 being the most feasible option when it comes to quick and suitable deployment in real-world eye disease classification tasks.

D. Grad-CAM Visualization

The GRAD-CAM visualization of the proposed model ResNet18 is shown in Fig. 9. Gradient-weighted Class Activation Mapping (Grad- CAM) was applied to interpret the prediction behavior of the highest- performing model, ResNet18, by visualizing the spatial regions that had the strongest impact on the predictions. As illustrated in the Grad- CAM visuals, the model repeatedly emphasized medically relevant ocular regions, especially the iris and adjacent corneal regions rather than irrelevant or background areas. For Uveitis cases, the heatmaps emphasize the central eye region, aligned with inflammation-prone re- gions. While Cataract samples show high attention over the lens area where lens opacity occurs. The alignment between high-activation regions and clinically recognized abnormalities indicates that the model captured meaningful disease-related features. This improves interpretability and supports the medical plausibility of the proposed system.

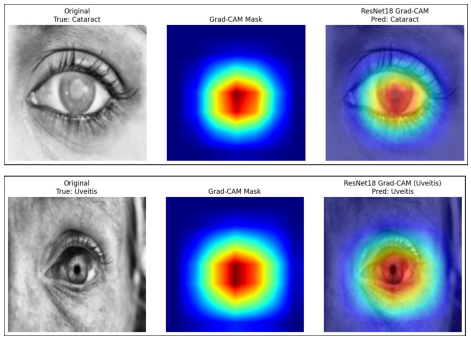


Fig. 9: Grad-CAM visualizations highlighting discriminative regions.

E. Model Comparison with Previous Study

Table IV shows a comparison between previous studies with our work. While other works primarily rely on OCT images, fundus im- ages or complex hybrid architectures, our proposed method performs disease

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classification by using simple eye images-a direction that is largely unexplored in the current literature. This study sets a baseline benchmark for future studies investigating eye disease classification using easily accessible non-OCT imaging data by achieving competitive accuracy with an optimized ResNet18 and reduced architectural complexity.

TABLE IV: Comparison with Previous Eye Disease Classification Studies

Study	Model	Test Acc. (%)	Remarks
[16]	Vision Transformer (ViT)	95.76	OCT-dependent; no statistical vali- dation
[17]	Mobile-ViT	99.17	High computational resources; lim- ited explainability
[18]	SViT (SqueezeNet + ViT)	90.27	Increased training cost due to hy- brid model
[19]	Two-Stream CNN-Transformer (LGSF-Net)	96.00	Complex architecture; lacks hyper- parameter tuning
Our Study	**ResNet18 (Optimized)**	**97.44**	**Strength: The proposed frame- work demonstrates high ac- curacy, fast training and in- ference, robust generalization across folds, and interpretable predictions via Grad-CAM.**

V. Conclusion

This study presented a robust deep learning framework combining CNNs and transformers for multi-class eye disease classification us- ing colored ocular images. Thorough experimental evaluation across six modern architectures revealed that ResNet18 yielded the best overall performance with a test accuracy of 97.44%, macro sensi- tivity of 95.11% and macro specificity of 99.36%, indicating strong consistency and generalization. Five-fold cross-validation additionally verified the robustness of the model with an average validation accuracy of approximately 96.8%. Grad-CAM based interpretability analysis revealed that model predictions were guided by clinically significant eye areas. Although the current system is restricted to a limited set of five ocular disease classes and cannot accommodate unseen or rare conditions. This confined phenomenon creates a scope in future to expand the work by including more disorder categories and multimodal clinical patient information. Overall, the framework demonstrates strong potential as a reliable and effective automated diagnosis system within its specific classes.

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